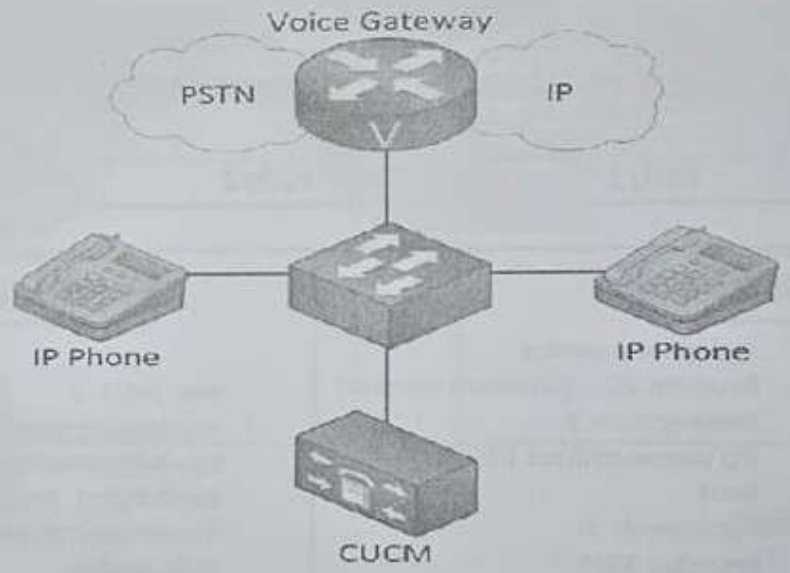
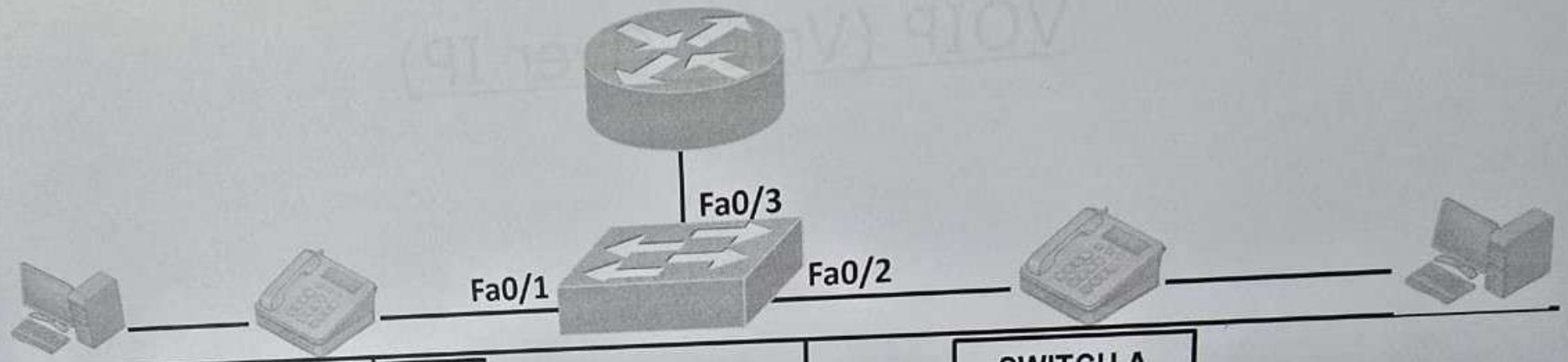
VOIP (Voice Over IP)



- ✓ *Voice Gateway: This is the router that is connected to our LAN/WAN but also to the PSTN (Public Switched Telephone Network)*
- ✓ *IP Phones: It needs IP address and will digitize your analog voice in IP packets that will be sent on the network. Most of the phones support POE (Power Over Ethernet) (#show power inline)*
- ✓ *CUCM (Cisco Unified Communication Manager): This is Cisco Call manager. It's where you configure dial plans, settings for IP phones. It's the equivalent of the PBX in old analog telephony system.*



ROUTER A

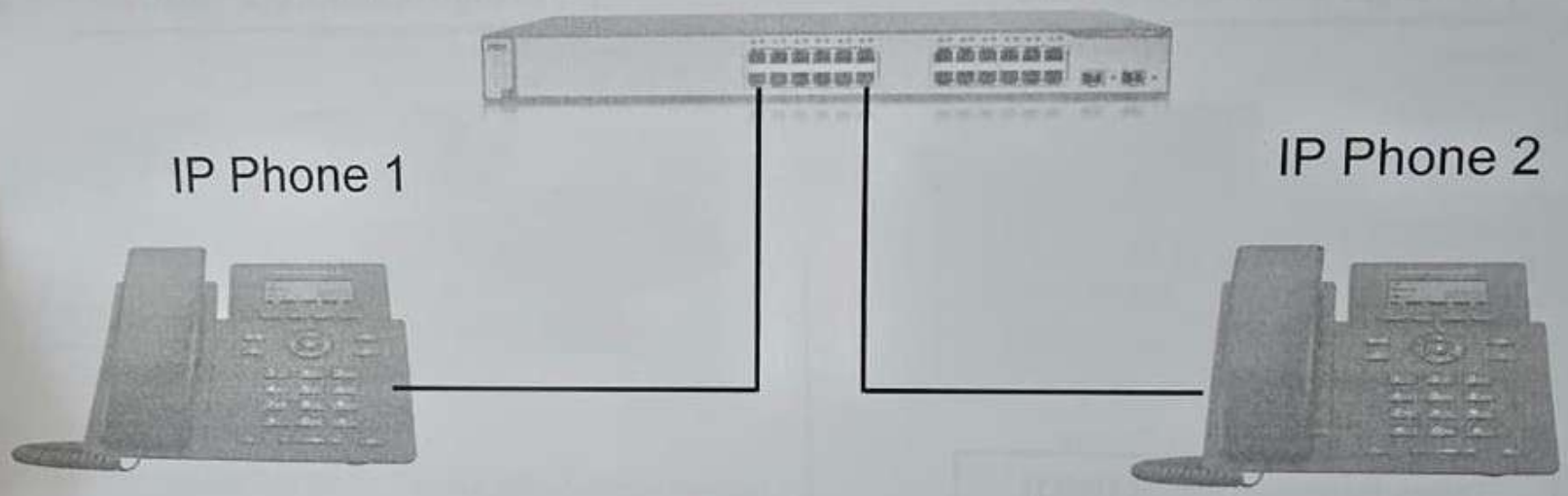
```
#int e0
#ip address 10.0.0.1 255.0.0.0
#exit
#ip dhcp pool systechvoip
#network 10.0.0.0 255.0.0.0
#default-router 10.0.0.1
#option 150 ip 10.0.0.1
#exit
#ip dhcp excluded-address 10.0.0.1
#sh ip dhcp binding
```

```
#telephony-service
#max-dn 10      (directory number)
#max-ephone 5
#ip sourse-address 10.0.0.1
#exit
#ephone-dn 1
#number 3001
#name cisco
#exit
#ephone-dn 2
#number 3002
#name ccnp
#exit
#ephone 1
#mac-address ****.****.****
#button 1:1
#exit
#ephone 2
#mac-address ****.****.****
#button 1:2
```

SWITCH A

```
#int  fa0/1- 3
#switchport mode access
#switchport access vlan 100
#switchport   voice vlan 200
#spanning-tree port fast
#cdp enable            (Cisco discovery protocol)
```

DHCP Configuration in PoE Switch



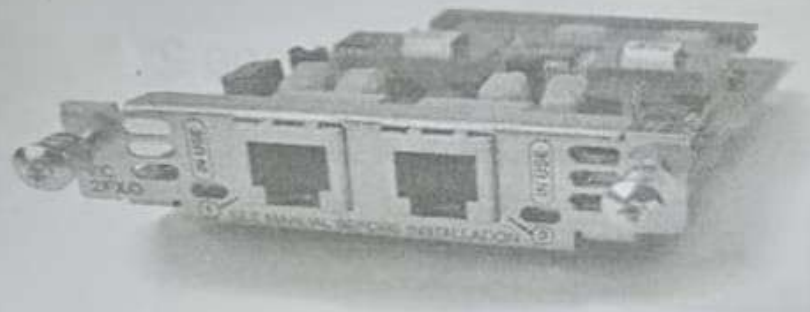
Ex: 192.168.3.100

Ex: 192.168.3.101

> Dial 192.168.3.101 from IP Phone 1

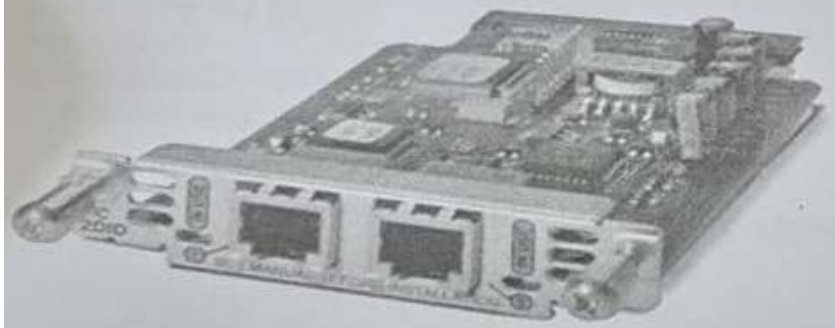
VOIP

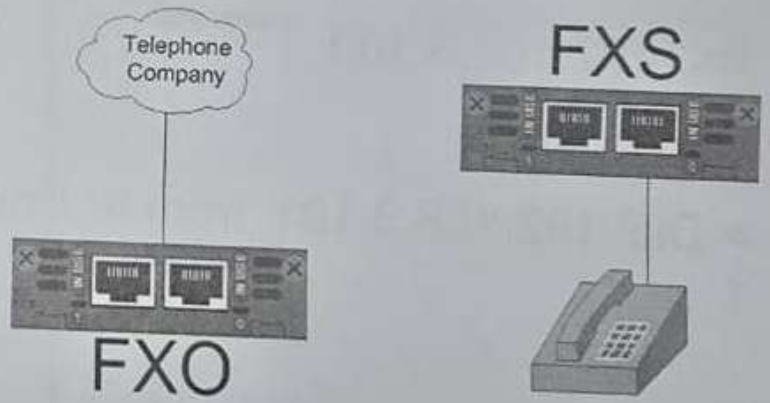
Foreign Exchange Office



Foreign-Exchange-Station

Direct-inward-dial (DID)





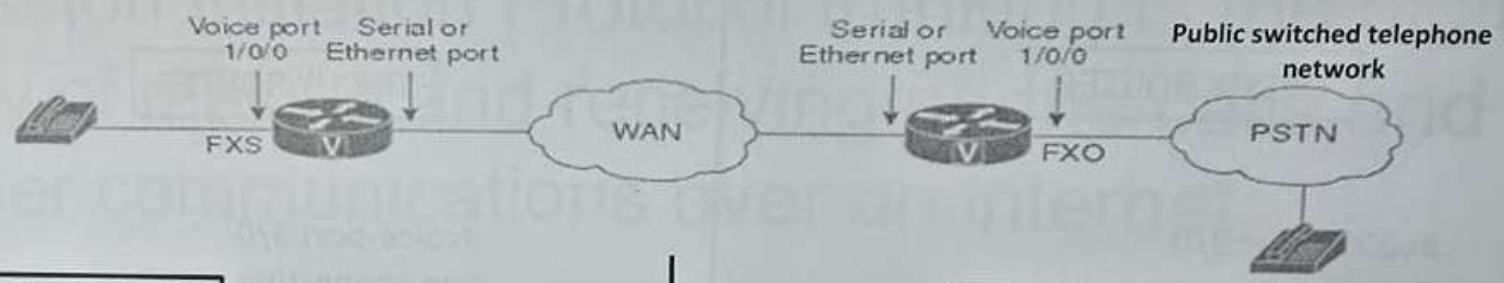
TRICHY
vi Block, St. Paul's Complex,Bharathiyar Salai,
Cantonment, Trichy - 620 001
75022 02555
contact@systechgroup.in

COIMBATORE
Ajay Building, 9th Street, Cross Cut Road,
Gandhipuram, Coimbatore - 641 012
75022 02555
systechgroup.in

CHENNAI
Thilakavathy Complex, Mudichur Road, Krishna Nagar,
West Tambaram, Chennai- 600 045
75022 02555
systechgroup

VOIP

FXS PORT CONFIGURATION



TRICHY ROUTER

```
#configure terminal
#dial-peer voice 10 pots    (plain old telephone service)
#destination-pattern 2001
#Port2/0
#exit
```

US ROUTER

```
#configure terminal
#dial-peer voice 10 pots
#destination-pattern 3001
#Port2/0
#exit
```

ROUTING VOICE

```
#dial-peer voice 20 voip
#destination-pattern 300.
#session target ipv4:20.0.0.2
#end
```

```
#dial-peer voice 20 voip
#destination-pattern 200.
#session target ipv4:20.0.0.1
#end
```

FXO PORT INCOMING CONFIGURATION

TRICHY ROUTER

```
#voice-port 3/0
#connection plar 2001
```

FXO PORT OUTGOING CONFIGURATION

TRICHY ROUTER

```
#dial-peer voice 2 pots
#destination-pattern 91
#Port 3/0
```

DIVERTING TRICHY CALL TO US

TRICHY ROUTER

```
#voice-port 3/0
#no connection plar 2001
#connection plar 3001
```

USING TRICHY LINE FROM US

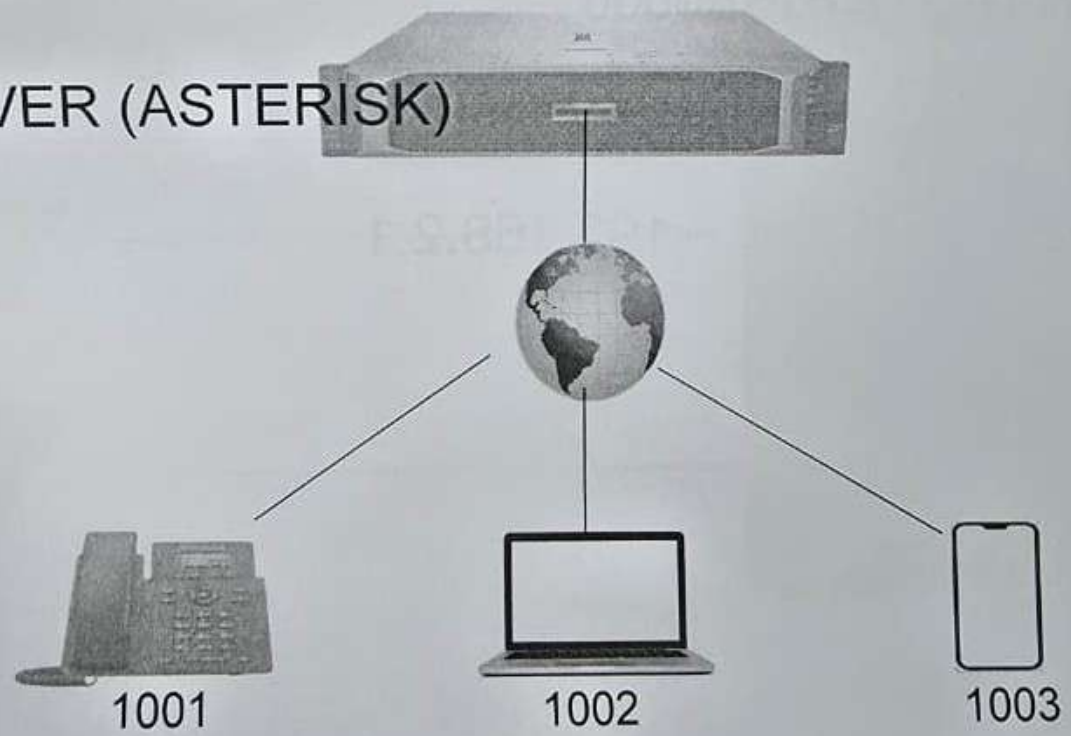
US ROUTER

```
#dial-peer voice 40 voip
#destination-pattern 91
#session target ipv4:20.0.0.1
#end
```

Session Initiation Protocol (SIP)

Session Initiation Protocol trunking is the digital way of making and receiving phone calls and other communications over an internet

SIP – SERVER (ASTERISK)



NAT TABLE

192.168.2.123:5060
192.168.2.123:5061
192.168.2.123:10000-20000

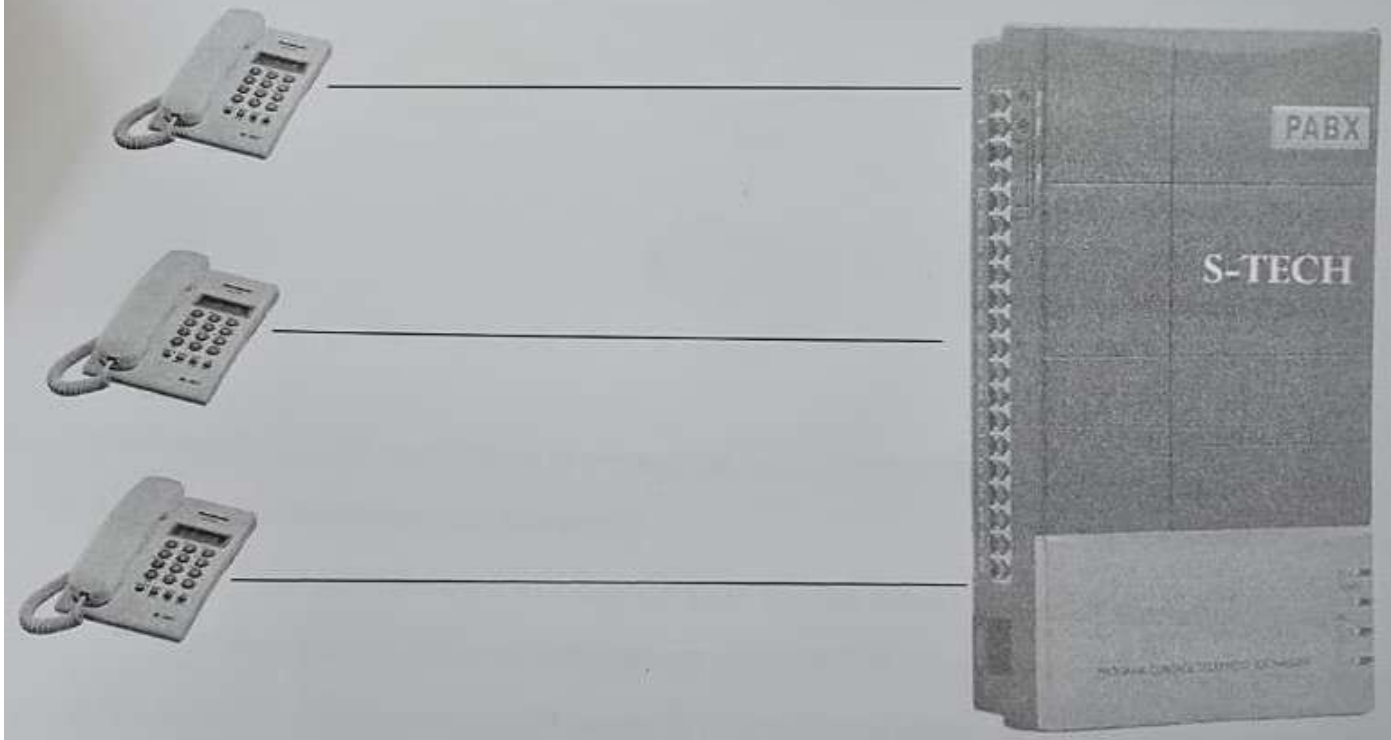
SIP – SERVER (ASTERISK)

192.168.2.123

192.168.2.1

Voice - Intercom

Private Automatic Branch
Exchange (PABX)



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